india-net

This is a look at the problems behind the sub par connectivity experience in India, and what can be done to resolve them.

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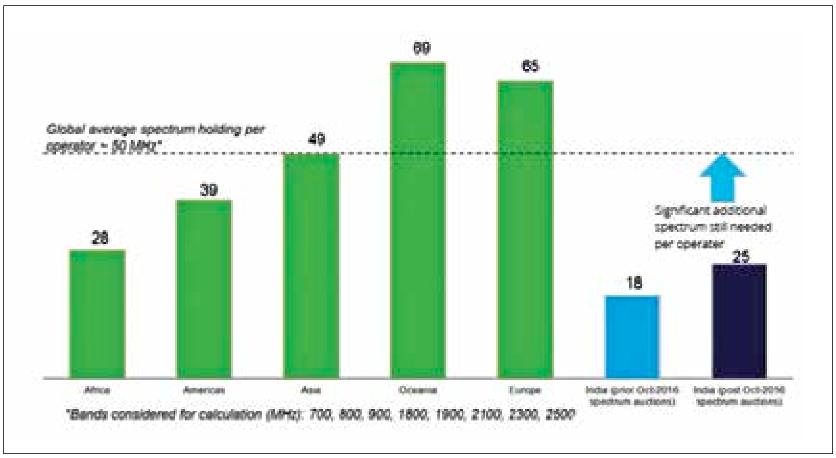
There are many companies and entities trying to fix the bad experience of the internet in India. These include the telecom operators, internet service providers, cable operators, think tanks, advocacy groups, as well as industry bodies such as COAI, IAMAI and GSMA. While the concerns of all these different organisations may be different, they all agree on the pain points discussed below, and the possible solutions.

Now the simplest, no brainer approach would be to just make it simple to lay cables. Let us imagine a cable passing through a hypothetical major railway station in a metropolitan city. Nearby, there is also a new subway station being built, as well as a junction between a national highway and a state highway. The junction is on the coast, near a forested area. This might sound like too many conditions, but it practically applies for most of Mumbai and its satellite cities. Now an operator would have to coordinate with a number of government agencies, including BMC, MMRDA, MMRC, MOEF, NHAI, MMB, MPCB, PWD and Railways. This is a nightmare situation because even after the cables are laid, any number of agencies can come and dig it up and accidentally cut the cables, which would then have to be put together again, a complicated procedure known as splicing. It is for this reason that no matter how much companies invest in the infrastructure, the quality is always terrible. The cabling in India has more splices per kilometre than recommended. It would simplify things for the service providers if there was a single window for seeking all the required permissions, which is provided in a timely manner. In Maharashtra, getting permissions involves a contract, and there is no stipulated time within which the agencies are required to give the necessary permissions. Right now, it is impossible for a service provider to realistically estimate how long it will take to get all the permissions to lay or maintain a cable, especially in busy cities. The funny and sad thing is that BSNL suffers from the red tape both internally and externally. One of the simplest ways in which this particular problem can be solved, is that the government can require ducts of a standardised specification to be built along major infrastructure projects, such as roads, bridges and railways, down to the more granular level of malls, stadiums, hospitals and housing complexes. This will allow widespread connectivity, which is also easily accessible for maintenance. The problem is that the government does not yet think of broadband access as an essential utility, such as water and electricity.

Another pain point is the internet exchange "service" provided by the government, through a company called the National Internet Exchange of India (NIXI). There are seven network operations centres located across the country, that provide switches for member networks to exchange data amongst themselves. So, a CDN provider, for example, can interface with an ISP, without having to route the traffic through various other networks. The membership fees is just too high for smaller ISPs, so they typically route their traffic through larger ISPs that connect to NIXI. Essentially, think of NIXI as a country-wide home network. Additionally, NIXI charges per GB of bandwidth, which makes the member networks feel the same pinch as you do when you think of FUP. The charges are ₹5 per GB of bandwidth, and all the ISPs want such charges to be done away with completely.

When it comes to wireless connectivity, spectrum availability is a problem. Compared to global levels, the amount of spectrum available to each operator is extremely low. Additionally, the allocated spectrum may not be contiguous, which means that operators have very little leeway when it comes to delivering services. While 2G or 3G can work well in a fractured spectrum, the more advanced technologies such as LTE and 5G require contiguous spectrum. In India, the DoT (for state broadcasting) and the MoD have reserved major portions of the commercially-viable spectrum, which is not even available to service providers. The switchover to digital television has opened up some free spectrum that was previously used for broadcast purposes.



Global average spectrum holding per Operator

Credit: Deloitte Touche Tohmatsu India LLP